RESUMO ÁREAS DE CONHECIMENTO SIPA: I-INOVAÇÃO

REFERÊNCIAS BÁSICAS I-INOVAÇÃO→OSLO MANUAL 2018 (OECD) E PESQUISA E DESENVOVIMENTO EXPERIMENTAL: FRASCATI MANUAL 2015 (OECD), E TECHNOLOGY READINESS LEVELS-TRLS: NASA.

JOÃO MELLO DA SILVA

EDISON ISHIKAWA

BRASÍLIA, OUTUBRO/2026

1-DESENVOLVIMENTO SUSTENTÁVEL: AGENDA 2030 DA ONU;

2-INOVAÇÃO: OSLO MANUAL 2018 (OECD) E PESQUISA E DESENVOVIMENTO EXPERIMENTAL: FRASCATI MANUAL 2015 (OECD), E TECHNOLOGY READINESS LEVELS-TRLS: NASA;

3-PRIVACIDADE (LEI GERAL DE DADOS PESSOAIS-LGPD, BRASIL, GENERAL DATA PROTECTION REGULATION-GDPR, UNIÃO EUROPEIA, FAIR INFORMATION PRACTICE PRINCIPLES-FIPPS, EUA); E

4-ANÁLISE DE NEGÓCIOS COM DECISÕES AJUSTADAS A RISCOS (PMI GUIDE TO BUSINESS ANALYSIS/2017, ISO 31.000/2009, ISO 31.000/2018, ISO 31.010/2019, PMBOK6/2017).

PREMISSA BÁSICA

"QUALQUER **SISTEMA SOCIAL** (**"ORGANIZAÇÃO**", SEGUNDO JAGUARIBE) COM **PROPÓSITO**, QUE **APRENDE E ENSINA**, CARACTERIZADO COMO **SIPA** (**S**USTENTÁVEL, **I**NOVADOR, **P**ROTETOR DA **P**RIVACIDADE, COM **A**NÁLISE DE NEGÓCIOS COM DECISÕES **A**JUSTADAS A RISCOS), DEVE TER COMO **PROPÓSITO** A **FORMAÇÃO "ÓTIMA" DE VALOR** (**VALORIZAÇÃO**) **E A APROPRIAÇÃO EQUÂNIME DO VALOR FORMADO** ÀS PARTES QUE PODEM AFETAR OU SER AFETADOS POR ESSA VALORIZAÇÃO (*STAKEHOLDERS*), SEJAM ELES INTERNOS OU EXTERNOS AO SISTEMA EM QUESTÃO, SEGUNDO **INDICADORES** DE **NÍVEIS** EM **TRAJETÓRIAS DE EXCELÊNCIA** (**EX1-**EFICIÊNCIA, **EX2-**EFICÁCIA, **EX3-**EMINÊNCIA, **EX4-**EFETIVIDADE) E DE **GRAUS** DE **EVOLUÇÃO ORGANIZACIONAL** (**EV1-**EXISTÊNCIA, **EV2-**EXPERIÊNCIA, **EV3-**EMERGÊNCIA, **EV4-**EVIDÊNCIA), **INDICADORES** ESSES ASSOCIADOS AOS **RESULTADOS** DE **TOMADA DE DECISÃO** (**ABSOLVER/reSOLVER/RESOLVER/DISSOLVER**) EM **MÚLTIPLOS NÍVEIS DE APRENDIZAGEM E ENSINO** (**FUNCIONAMENTO/PREMISSAS/INTELIGÊNCIA/CONTEXTOS**)".

(i) **17 ODS E AS 169 CORRESPONDENTES METAS DA 2030 AGENDA/UN**, (https://sustainabledevelopment.un.org/content/documents/21252030%20Agenda%20for%20Sustainable%20Development%20web.pdf);

(ii) **CONCEITOS DE INOVAÇÃO EM PRODUTOS, PROCESSOS E PRONTIDÃO DO OSLO MANUAL 2018/OECD** (http://www.oecd.org/science/oslo-manual-2018-9789264304604-en.htm), **CONCEITOS DE PESQUISA E DESENVOLVIMENTO EXPERIMENTAL DO FRASCATI MANUAL 2015/OECD** (https://www.oecd.org/en/publications/2015/10/frascati-manual-2015_g1g57dcb.html), **9 TECHNOLOGY READINESS LEVELS (TRLS) DA NASA** (https://www.nasa.gov/pdf/458490main_TRL_Definitions.pdf), **CIBERSEGURANÇA (CYBERSECURITY), COM ATENÇÃO NA EDUCAÇÃO EM CYBERSECURITY/US** (https://www.nist.gov/itl/applied-cybersecurity/nice);

(iii) **DISPOSITIVOS LEGAIS E SUAS REGULAMENTAÇÕES QUANTO À PROTEÇÃO DE DADOS PESSOAIS (LGPD/BRASIL:** HTTP://WWW.PLANALTO.GOV.BR/CCIVIL_03/_ATO2019-2022/2019/LEI/L13853.HTM, **GDPR/EU:** https://gdpr.eu/what-is-gdpr/**, PRIVACY ENGINEERING/US:** https://nvlpubs.nist.gov/nistpubs/ir/2017/NIST.IR.8062.pdf, https://blog.lukaszolejnik.com/privacy-engineering-principles-recommended-by-nist/);

(iv) **CONCEITOS BÁSICOS DE ANÁLISE DE NEGÓCIOS, SEGUNDO O PMI GUIDE TO BUSINESS ANALYSIS 2017 E DE GESTÃO DE RISCOS EM NORMAS ISO (ISO 31.000/2009, ISO 31.000/2018, ISO 31.010/2019)**.

PRONTIDÃO EM INOVAÇÃO

AS REFERÊNCIAS BÁSICAS SÃO O OSLO MANUAL 2018 E OS TECHNOLOGY READINESS LEVELS-TRLS, DA NASA.

OSLO MANUAL 2018

O OSLO MANUAL 2018, PUBLICADO PELA ORGANIZATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT – OECD, TRATA INOVAÇÃO EM PRODUTO, PROCESSO E PRONTIDÃO (http://www.oecd.org/science/oslo-manual-2018-9789264304604-en.htm).

CONFORME O ITEM 1.3.1 NATURE OF THE GUIDANCE IN THIS MANUAL, PÁGINA 39

"THE PURPOSE OF THIS MANUAL IS TO GUIDE INNOVATION DATA COLLECTION AND REPORTING EFFORTS THROUGH A COMMON VOCABULARY, AGREED PRINCIPLES AND PRACTICAL CONVENTIONS. THESE CAN ENHANCE THE COMPARABILITY OF STATISTICAL OUTPUTS AND SUPPORT THE PROGRESSIVE DEVELOPMENT OF A GLOBAL STATISTICAL INFORMATION INFRASTRUCTURE ON INNOVATION THAT IS RELEVANT AND USEFUL FOR RESEARCHERS AND DECISION MAKERS."

RECOMENDA-SE A LEITURA INICIAL DO GLOSSÁRIO PARA IDENTIFICAR POSSÍVEIS TÓPICOS DE INTERESSE PARA OS PROJETOS. VISANDO VERIFICAR O ENTENDIMENTO BÁSICO SOBRE INOVAÇÃO (INNOVATION) NO ÂMBITO DO OSLO MANUAL 2018, OS TEXTOS LISTADOS NO GLOSSÁRIO (GLOSSARY), PÁGINAS 241 A 254 DO OSLO MANUAL 2018, SERÃO ALOCADOS A PERGUNTAS TIPO MÚLTIPLA ESCOLHA E VERDADEIRA/FALSA EM QUESTIONÁRIOS ONLINE SOBRE INOVAÇÃO.

TRANSCRIÇÃO DE ALGUNS TRECHOS OSLO MANUAL 2018

“BUSINESS MODEL INNOVATION RELATES TO CHANGES IN A FIRM’S CORE BUSINESS PROCESSES AS WELL AS IN THE MAIN PRODUCTS IT SELLS, CURRENTLY AND IN THE FUTURE.” (PAGE 242)

“A KEY TENET OF THE OSLO MANUAL IS THAT INNOVATION CAN AND SHOULD BE MEASURED. THE REQUIREMENTS FOR MEASURABILITY IS AN ESSENTIAL CRITERION FOR SELECTING THE CONCEPTS, DEFINITIONS AND CLASSIFICATIONS IN THIS MANUAL. [...] KEY COMPONENTS OF THE CONCEPT OF INNOVATION INCLUDE THE ROLE OF KNOWLEDGE AS A BASIS FOR INNOVATION, NOVELTY AND UTILITY, AND VALUE CREATION OR PRESERVATION AS THE PRESUMED GOAL OF INNOVATION. THE REQUIREMENT FOR IMPLEMENTATION DIFFERENTIATES INNOVATION FROM OTHER CONCEPTS SUCH AS INVENTION, AS AN INNOVATION MUST BE IMPLEMENTED, I.E. PUT INTO USE OR MADE AVAILABLE FOR OTHERS TO USE.

THE TERM “INNOVATION” CAN SIGNIFY BOTH AN ACTIVITY AND THE OUTCOME OF AN ACTIVITY. [...] AN INNOVATION IS A NEW OR IMPROVED PRODUCT OR PROCESS (OR COMBINATION THEREOF) THAT DIFFERS SIGNIFICANTLY FROM THE UNIT’S PREVIOUS PRODUCTS OR PROCESSES AND THAT HAS BEEN MADE AVAILABLE TO POTENTIAL USERS (PRODUCT) OR BROUGHT INTO USE BY THE UNIT (PROCESS). THIS DEFINITION USES THE GENERIC TERM “UNIT” TO DESCRIBE THE ACTOR RESPONSIBLE FOR INNOVATIONS. IT REFERS TO ANY INSTITUTIONAL UNIT IN ANY SECTOR, INCLUDING HOUSEHOLDS AND THEIR INDIVIDUAL MEMBERS. THIS DEFINITION IS FURTHER DEVELOPED AND OPERATIONALIZED TO PROVIDE THE BASIS FOR THE PRACTICAL GUIDELINES IN THIS MANUAL FOR THE BUSINESS SECTOR. ...” (PAGE 20)

TRANSCRIÇÃO DE ALGUNS TRECHOS OSLO MANUAL 2018

INNOVATION ACTIVITIES INCLUDE ALL DEVELOPMENTAL, FINANCIAL AND COMMERCIAL ACTIVITIES UNDERTAKEN BY A FIRM THAT ARE INTENDED TO RESULT IN AN INNOVATION FOR THE FIRM. A BUSINESS INNOVATION IS A NEW OR IMPROVED PRODUCT OR BUSINESS PROCESS (OR COMBINATION THEREOF) THAT DIFFERS SIGNIFICANTLY FROM THE FIRM'S PREVIOUS PRODUCTS OR BUSINESS PROCESSES AND THAT HAS BEEN INTRODUCED ON THE MARKET OR BROUGHT INTO USE BY THE FIRM. COMPARED TO THE PREVIOUS EDITION, A MAJOR CHANGE FOR THE DEFINITION OF BUSINESS INNOVATION [...] HAS BEEN THE REDUCTION, [...], IN THE COMPLEXITY OF THE PREVIOUS LIST-BASED DEFINITION OF FOUR TYPES OF INNOVATIONS (PRODUCT, PROCESS, ORGANIZATIONAL AND MARKETING), TO TWO MAIN TYPES: PRODUCT INNOVATIONS AND BUSINESS PROCESS INNOVATIONS. THE REVISED EDITION ALSO REDUCES THE AMBIGUITY OF THE REQUIREMENTS FOR A "SIGNIFICANT" CHANGE BY COMPARING BOTH NEW AND IMPROVED INNOVATIONS TO THE FIRM'S EXISTING PRODUCTS OR BUSINESS PROCESSES. [...] A PRODUCT INNOVATION IS A NEW OR IMPROVED GOOD OR SERVICE THAT DIFFERS SIGNIFICANTLY FROM THE FIRM'S PREVIOUS GOODS OR SERVICES AND THAT HAS BEEN INTRODUCED ON THE MARKET. A BUSINESS PROCESS INNOVATION IS A NEW OR IMPROVED BUSINESS PROCESS FOR ONE OR MORE BUSINESS FUNCTIONS THAT DIFFERS SIGNIFICANTLY FROM THE FIRM'S PREVIOUS BUSINESS PROCESSES AND THAT HAS BEEN BROUGHT INTO USE BY THE FIRM. BUSINESS PROCESS INNOVATIONS CONCERN SIX DIFFERENT FUNCTIONS OF A FIRM, AS IDENTIFIED IN THE BUSINESS MANAGEMENT LITERATURE. [...] THE TAXONOMY OF BUSINESS FUNCTIONS PROPOSED IN THIS MANUAL MAPS REASONABLY WELL ONTO THE PREVIOUS EDITION'S CATEGORIES OF PROCESS, MARKETING AND ORGANIZATIONAL INNOVATIONS." (PAGES 20/21

TRANSCRIÇÃO DE ALGUNS TRECHOS OSLO MANUAL 2018

"BUSINESS INNOVATION ACTIVITIES INCLUDE ALL DEVELOPMENTAL, FINANCIAL AND COMMERCIAL ACTIVITIES UNDERTAKEN BY A FIRM THAT ARE INTENDED TO RESULT IN AN INNOVATION FOR THE FIRM. THEY INCUDE:

- **RESEARCH AND EXPERIMENTAL DEVELOPMENT (R&D) ACTIVITIES.**
- **ENGINEERING, DESIGN AND OTHER CREATIVE WORK ACTIVITIES.**
- **MARKETING AND BRAND EQUITY ACTIVITIES.**
- **INTELLECTUAL PROPERTY (IP) RELATED ACTIVITIES.**
- **EMPLOYEE TRAINING ACTIVITIES.**
- **SOFTWARE DEVELOPMENT AND DATABASE ACTIVITIES.**
- **ACTIVITIES RELATED TO THE ACQUISITION OR LEASE OF TANGIBLE ASSETS.**
- **INNOVATION MANAGEMENT ACTIVITIES.**

INNOVATION ACTIVITIES CAN RESULT IN AN INNOVATION, BE ONGOING, POSTPONED OR ABANDONED. AN INNOVATION INDICATOR IS A STATISTICAL SUMMARY MEASURE OF AN INNOVATION PHENOMENON (ACTIVITY, OUTPUT, EXPENDITURE, ETC.) OBSERVED IN A POPULATION OR A SAMPLE THERE OF FOR A SPECIFIED TIME OR PLACE. INDICATORS ARE USUALLY CORRECTED (OR STANDARDIZED) TO PERMIT COMPARISONS ACROSS UNITS THAT DIFFER IN SIZE OR OTHER CHARACTERISTICS. INNOVATION OBJECTIVES CONSIST OF A FIRM'S IDENTIFIABLE GOALS, THAT REFLECT ITS MOTIVES AND UNDERLYING STRATEGIES WITH RESPECT TO ITS INNOVATION EFFORTS. THE OBJECTIVES CAN CONCERN THE CHARACTERISTICS OF THE INNOVATION ITSELF, SUCH AS ITS SPECIFICATIONS, OR ITS MARKET AND ECONOMIC OBJECTIVES. INNOVATION OUTCOMES ARE THE OBSERVED EFFECTS OF INNOVATIONS, INCLUDING THE EXTENT TO WHICH A FIRM'S OBJECTIVES ARE MET AND THE BROADER EFFECTS OF INNOVATION ON OTHER ORGANIZATIONS, THE ECONOMY, SOCIETY, AND THE ENVIRONMENT." (PAGE 247)

MANUAL FRASCATI 2015

COM FOCO NO **FRASCATI MANUAL 2015, GUIDELINES FOR COLLECTING AND REPORTING DATA ON RESEARCH AND EXPERIMENTAL DEVELOPMENT**, **THE MEASUREMENT OF SCIENTIFIC, TECHNOLOGICAL AND INNOVATION ACTIVITIES**, **OECD PUBLISHING**, **PARIS**, (**https://www.oecd.org/en/publications/2015/10/frascati-manual-2015_g1g57dcb.html**), PODE-SE CONSOLIDAR **PESQUISA E DESENVOLVIMENTO EXPERIMENTAL=RESEARCH AND EXPERIMENTAL DEVELOPMENT=RED** EM TERMOS DE **PESQUISA E DESENVOLVIMENTO EXPERIMENTAL EM PRODUTOS/RESEARCH AND EXPERIMENTAL DEVELOPMENT IN PRODUCTS=PDR=RED1**; **PESQUISA E DESENVOLVIMENTO EXPERIMENTAL EM PROCESSOS/RESEARCH AND EXPERIMENTAL DEVELOPMENT IN PROCESSES=PPR=RED2**; E **COLABORAÇÃO EM PESQUISA E DESENVOLVIMENTO/RESEARCH AND EXPERIMENTAL DEVELOPMENT COLLABORATION=COR=RED3**.

NASA TECHNOLOGY READINESS LEVELS-TRLS

EM UMA ADAPTAÇÃO DA TABELA "TECHNOLOGY READINESS LEVEL DEFINITIONS" (https://www.nasa.gov/pdf/458490main_trl_definitions.pdf), OS 9 NÍVEIS DE PRONTIDÃO TECNOLÓGICA/TECHNOLOGY READINESS LEVELS – TRLS DA NASA PODEM SER COMBINADOS EM 6 CONCENTRAÇÕES (A/B/C/D/E/F).

ESSAS CONCENTRAÇÕES DELIMITAM OS PRINCIPAIS PARTICIPANTES: 1-ACADEMIA/ACADEMIA; 2-GOVERNO E ENTIDADES REPRESENTATIVAS DA SOCIEDADE CIVIL/GOVERNMENT AND CIVIL SOCIETY REPRESENTATIVE ENTITIES; AND 3-SETOR PRIVADO, ENTIDADES ESTATAIS E ONGS/PRIVATE SECTOR, STATE ENTITIES AND NGOs

RESPECTIVOS PAPÉIS: 1-CRIAÇÃO DE CONHECIMENTO/KNOWLEDGE CREATION; 2-INCENTIVO E SUPORTE À COLABORAÇÃO/INCENTIVE AND SUPPORT TO COLLABORATION; AND 3-PRODUTOS COMERCIAIS E SUCESSO NA IMPLEMENTAÇÃO DE SERVIÇOS PÚBLICOS/COMMERCIAL PRODUCTS AND PUBLIC SERVICES IMPLEMENTATION SUCCESS).

É POSSÍVEL GERAR DEFINIÇÕES OPERACIONAIS EM INGLÊS PARA OS PAPÉIS DOS PRINCIPAIS PARTICIPANTES EM TERMOS DE SEUS COMPONENTS, QUE SERVEM DE BASE PARA CARACTERIZAÇÃO DAS ATIVIDADES RELACIONADAS

TECHNOLOGY READINESS LEVELS (Reference: https://www.nasa.gov/pdf/458490main_TRL_Definitions.pdf)

TRL	DEFINITION	HARDWARE DESCRIPTION	SOFTWARE DESCRIPTION	EXIT CRITERIA
1	BASIC PRINCIPLES OBSERVED AND REPORTED	SCIENTIFIC KNOWLEDGE GENERATED UNDERPINNING HARDWARE TECHNOLOGY CONCEPTS/APPLICATIONS.	SCIENTIFIC KNOWLEDGE GENERATED UNDERPINNING BASIC PROPERTIES OF SOFTWARE ARCHITECTURE AND MATHEMATICAL FORMULATION.	PEER REVIEWED PUBLICATION OF RESEARCH UNDERLYING THE PROPOSED CONCEPT/APPLICATION
2	TECHNOLOGY CONCEPT AND/OR APPLICATION	INVENTION BEGINS, PRACTICAL APPLICATION IS IDENTIFIED BUT IS SPECULATIVE, NO EXPERIMENTAL PROOF OR DETAILED ANALYSIS IS AVAILABLE TO SUPPORT THE CONJECTURE.	PRACTICAL APPLICATION IS IDENTIFIED BUT IS SPECULATIVE, NO EXPERIMENTAL PROOF OR DETAILED ANALYSIS IS AVAILABLE TO SUPPORT THE CONJECTURE. BASIC PROPERTIES OF ALGORITHMS, REPRESENTATIONS AND CONCEPTS DEFINED. BASIC PRINCIPLES CODED. EXPERIMENTS PERFORMED WITH SYNTHETIC DATA.	DOCUMENTED DESCRIPTION OF THE APPLICATION/CONCEPT THAT ADDRESSES FEASIBILITY AND BENEFIT.
3	ANALYTICAL AND EXPERIMENTAL CRITICAL FUNCTION AND/OR CHARACTERISTIC PROOF OF CONCEPT	ANALYTICAL STUDIES PLACE THE TECHNOLOGY IN AN APPROPRIATE CONTEXT AND LABORATORY DEMONSTRATIONS, MODELING AND SIMULATION VALIDATE ANALYTICAL PREDICTION.	DEVELOPMENT OF LIMITED FUNCTIONALITY TO VALIDATE CRITICAL PROPERTIES AND PREDICTIONS USING NON-INTEGRATED SOFTWARE COMPONENTS.	DOCUMENTED ANALYTICAL/EXPERIMENTAL RESULTS VALIDATING PREDICTIONS OF KEY PARAMETERS
4	COMPONENT AND/OR BREADBOARD VALIDATION IN LABORATORY ENVIRONMENT	A LOW FIDELITY SYSTEM/COMPONENT BREADBOARD IS BUILT AND OPERATED TO DEMONSTRATE BASIC FUNCTIONALITY AND CRITICAL TEST ENVIRONMENTS, AND ASSOCIATED PERFORMANCE PREDICTIONS ARE DEFINED RELATIVE TO THE FINAL OPERATING ENVIRONMENT.	KEY, FUNCTIONALLY CRITICAL, SOFTWARE COMPONENTS ARE INTEGRATED, AND FUNCTIONALLY VALIDATED, TO ESTABLISH INTEROPERABILITY AND BEGIN ARCHITECTURE DEVELOPMENT. ENVIRONMENT DEFINED AND PERFORMANCE IN THIS ENVIRONMENT PREDICTED.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS. DOCUMENTED DEFINITION OF RELEVANT ENVIRONMENT.
5	COMPONENT AND/OR BREADBOARD VALIDATION IN RELEVANT ENVIRONMENT	A MEDIUM FIDELITY SYSTEM/COMPONENT BRASSBOARD IS BUILT AND OPERATED TO DEMONSTRATE OVERALL PERFORMANCE IN A SIMULATED OPERATIONAL ENVIRONMENT WITH REALISTIC SUPPORT ELEMENTS THAT DEMONSTRATES OVERALL PERFORMANCE IN CRITICAL AREAS. PERFORMANCE PREDICTIONS ARE MADE FOR SUBSEQUENT DEVELOPMENT PHASES.	END-TO-END SOFTWARE ELEMENTS IMPLEMENTED AND INTERFACED WITH EXISTING SYSTEMS/SIMULATIONS CONFORMING TO TARGET ENVIRONMENT. END-TO-END SOFTWARE SYSTEM, TESTED IN RELEVANT ENVIRONMENT, MEETING PREDICTED PERFORMANCE. OPERATIONAL ENVIRONMENT PERFORMANCE PREDICTED. PROTOTYPE IMPLEMENTATIONS DEVELOPED.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS. DOCUMENTED DEFINITION OF SCALING REQUIREMENTS.
6	SYSTEM/SUBSYSTEM MODEL OR PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT	A HIGH FIDELITY SYSTEM/COMPONENT PROTOTYPE THAT ADEQUATELY ADDRESSES ALL CRITICAL SCALING ISSUES IS BUILT AND OPERATED IN A RELEVANT ENVIRONMENT TO DEMONSTRATE OPERATIONS UNDER CRITICAL ENVIRONMENTAL CONDITIONS.	PROTOTYPE IMPLEMENTATION OF THE SOFTWARE DEMONSTRATED ON FULL-SCALE REALISTIC PROBLEMS. PARTIALLY INTEGRATED WITH EXISTING HARDWARE/SOFTWARE SYSTEMS. LIMITED DOCUMENTATION AVAILABLE. ENGINEERING FEASIBILITY FULLY DEMONSTRATED.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS.
7	SYSTEM PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT	A HIGH FIDELITY ENGINEERING UNIT THAT ADEQUATELY ADDRESSES ALL CRITICAL SCALING ISSUES IS BUILT AND OPERATED IN A RELEVANT ENVIRONMENT TO DEMONSTRATE PERFORMANCE IN THE ACTUAL OPERATIONAL ENVIRONMENT AND PLATFORM (GROUND, AIRBORNE, OR SPACE).	PROTOTYPE SOFTWARE EXISTS HAVING ALL KEY FUNCTIONALITY AVAILABLE FOR DEMONSTRATION AND TEST. WELL INTEGRATED WITH OPERATIONAL HARDWARE/SOFTWARE SYSTEMS DEMONSTRATING OPERATIONAL FEASIBILITY. MOST SOFTWARE BUGS REMOVED. LIMITED DOCUMENTATION AVAILABLE.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS.
8	ACTUAL SYSTEM COMPLETED AND "FLIGHT QUALIFIED" THROUGH TEST AND DEMONSTRATION	THE FINAL PRODUCT IN ITS FINAL CONFIGURATION IS SUCCESSFULLY DEMONSTRATED THROUGH TEST AND ANALYSIS FOR ITS INTENDED OPERATIONAL ENVIRONMENT AND PLATAFORM (GROUND, AIRBORNE, OR SPACE).	ALL SOFTWARE HAS BEEN THOROUGHLY DEBUGGED AND FULLY INTEGRATED WITH ALL OPERATIONAL HARDWARE AND SOFTWARE SYSTEMS. ALL USER DOCUMENTATION, TRAINING DOCUMENTATION, AND MAINTENANCE DOCUMENTATION COMPLETED. ALL FUNCTIONALITY SUCCESSFULLY DEMONSTRATED IN SIMULATED OPERATIONAL SCENARIOS. VERIFICATION AND VALIDATION (V&V) COMPLETED.	DOCUMENTED TEST PERFORMANCE VERIFYING AGREEMENT WITH ANALYTICAL PREDICTIONS.
9	ACTUAL SYSTEM FLIGHT PROVEN THROUGH SUCCESSFUL MISSION OPERATIONS	THE FINAL PRODUCT IS SUCCESSFULLY OPERATED IN AN ACTUAL MISSION.	ALL SOFTWARE HAS BEEN THOROUGHLY DEBUGGED AND FULLY INTEGRATED WITH ALL OPERATIONAL HARDWARE AND SOFTWARE SYSTEMS. ALL DOCUMENTATION HAS BEEN COMPLETED. SUSTAINING SOFTWARE ENGINEERING SUPPORT IS IN PLACE. SYSTEM HAS BEEN SUCCESSFULLY OPERATED IN THE OPERATIONAL ENVIRONMENT.	DOCUMENTED MISSION OPERATIONAL RESULTS.

MAJOR PLAYERS AND ROLES		
ACADEMIA	GOVERNMENT & CIVIL SOCIETY REPRESENTATIVE ENTITIES	PRIVATE SECTOR STATE ENTITIES NGOs
KNOWLEDGE CREATION	INCENTIVE & SUPPORT TO COLLABORATION	COMMERCIAL PRODUCTS & PUBLIC SERVICES IMPLEMENTATION SUCCESS

CONCENTRATIONS OF TRLs					
1, 2, 3	2, 3, 4	3, 4, 5	5, 6	6, 7, 8	7, 8, 9
A	B	C	D	E	F

A-BASIC TECHNOLOGY RESEARCH TRLs 1, 2, 3

B-RESEARCH TO PROVE FEASIBILITY TRLs 2, 3, 4

C-TECHNOLOGY DEMONSTRATION TRLs 3, 4, 5

D-TECHNOLOGY DEVELOPMENT AND PROTOTYPES TRLs 5, 6

E-PILOT PLAN AND SCALE UP TRLs 6, 7, 8

F-MARKET LAUNCH AND COMMERCIALIZATION TRLs 7, 8, 9

TECHNOLOGY READINESS LEVELS (Reference: https://www.nasa.gov/pdf/458490main_TRL_Definitions.pdf)

TRL	DEFINITION	HARDWARE DESCRIPTION	SOFTWARE DESCRIPTION	EXIT CRITERIA	MAJOR PLAYERS AND ROLES: ACADEMIA — KNOWLEDGE CREATION	MAJOR PLAYERS AND ROLES: GOVERNMENT & CIVIL SOCIETY REPRESENTATIVE ENTITIES — INCENTIVE & SUPPORT TO COLLABORATION	MAJOR PLAYERS AND ROLES: PRIVATE SECTOR STATE ENTITIES NGOs — COMMERCIAL PRODUCTS & PUBLIC SERVICES IMPLEMENTATION SUCCESS
					CONCENTRATIONS OF TRLs		
					1, 2, 3 (A)	2, 3, 4 (B)	3, 4, 5 (C)
							5, 6 (D)
							6, 7, 8 (E)
							7, 8, 9 (F)
1	BASIC PRINCIPLES OBSERVED AND REPORTED	SCIENTIFIC KNOWLEDGE GENERATED UNDERPINNING HARDWARE TECHNOLOGY CONCEPTS/APPLICATIONS.	SCIENTIFIC KNOWLEDGE GENERATED UNDERPINNING BASIC PROPERTIES OF SOFTWARE ARCHITECTURE AND MATHEMATICAL FORMULATION.	PEER REVIEWED PUBLICATION OF RESEARCH UNDERLYING THE PROPOSED CONCEPT/APPLICATION.	A-BASIC TECHNOLOGY RESEARCH TRLs 1, 2, 3		
2	TECHNOLOGY CONCEPT AND/OR APPLICATION	INVENTION BEGINS, PRACTICAL APPLICATION IS IDENTIFIED BUT IS SPECULATIVE, NO EXPERIMENTAL PROOF OR DETAILED ANALYSIS IS AVAILABLE TO SUPPORT THE CONJECTURE.	PRACTICAL APPLICATION IS IDENTIFIED BUT IS SPECULATIVE, NO EXPERIMENTAL PROOF OR DETAILED ANALYSIS IS AVAILABLE TO SUPPORT THE CONJECTURE. BASIC PROPERTIES OF ALGORITHMS, REPRESENTATIONS AND CONCEPTS DEFINED. BASIC PRINCIPLES CODED. EXPERIMENTS PERFORMED WITH SYNTHETIC DATA.	DOCUMENTED DESCRIPTION OF THE APPLICATION/CONCEPT THAT ADDRESSES FEASIBILITY AND BENEFIT.		B-RESEARCH TO PROVE FEASIBILITY TRLs 2, 3, 4	
3	ANALYTICAL AND EXPERIMENTAL CRITICAL FUNCTION AND/OR CHARACTERISTIC PROOF OF CONCEPT	ANALYTICAL STUDIES PLACE THE TECHNOLOGY IN AN APPROPRIATE CONTEXT AND LABORATORY DEMONSTRATIONS, MODELING AND SIMULATION VALIDATE ANALYTICAL PREDICTION.	DEVELOPMENT OF LIMITED FUNCIONALITY TO VALIDADTE CRITICAL PROPERTIES AND PREDICTIONS USING NON-INTEGRATED SOFTWARE COMPONENTS.	DOCUMENTED ANALYTICAL/EXPERIMENTAL RESULTS VALIDATING PREDICTIONS OF KEY PARAMETERS.			C-TECHNOLOGY DEMONSTRATION TRLs 3, 4, 5
4	COMPONENT AND/OR BREADBOARD VALIDATION IN LABORATORY ENVIRONMENT	A LOW FIDELITY SYSTEM/COMPONENT BREADBOARD IS BUILT AND OPERATED TO DEMONSTRATE BASIC FUNCTIONALITY AND CRITICAL TEST ENVIRONMENTS, AND ASSOCIATED PERFORMANCE PREDICTIONS ARE DEFINED RELATIVE TO THE FINAL OPERATING ENVIRONMENT.	KEY, FUNCTIONALLY CRITICAL, SOFTWARE COMPONENTS ARE INTEGRATED, AND FUNCTIONALLY VALIDATED, TO ESTABLISH INTEROPERABILITY AND BEGIN ARCHITECTURE DEVELOPMENT. ENVIRONMENT DEFINED AND PERFORMANCE IN THIS ENVIRONMENT PREDICTED.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS. DOCUMENTED DEFINITION OF RELEVANT ENVIRONMENT.			
5	COMPONENT AND/OR BREADBOARD VALIDATION IN RELEVANT ENVIRONMENT	A MEDIUM FIDELITY SYSTEM/COMPONENT BRASSBOARD IS BUILT AND OPERATED TO DEMONSTRATE OVERALL PERFORMANCE IN A SIMULATED OPERATIONAL ENVIRONMENT WITH REALISTIC SUPPORT ELEMENTS THAT DEMONSTRATES OVERALL PERFORMANCE IN CRITICAL AREAS. PERFORMANCE PREDICTIONS ARE MADE FOR SUBSEQUENT DEVELOPMENT PHASES.	END-TO-END SOFTWARE ELEMENTS IMPLEMENTED AND INTERFACED WITH EXISTING SYSTEMS/SIMULATIONS CONFORMING TO TARGET ENVIRONMENT. END-TO-END SOFTWARE SYSTEM, TESTED IN RELEVANT ENVIRONMENT, MEETING PREDICTED PERFORMANCE. OPERATIONAL ENVIRONMENT PERFORMANCE PREDICTED. PROTOTYPE IMPLEMENTATIONS DEVELOPED.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS. DOCUMENTED DEFINITION OF SCALING REQUIREMENTS.			
6	SYSTEM/SUB-SYSTEM MODEL OR PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT	A HIGH FIDELITY SYSTEM/COMPONENT PROTOTYPE THAT ADEQUATELY ADDRESSES ALL CRITICAL SCALING ISSUES IS BUILT AND OPERATED IN A RELEVANT ENVIRONMENT TO DEMONSTRATE OPERATIONS UNDER CRITICAL ENVIRONMENTAL CONDITIONS.	PROTOTYPE IMPLEMENTATION OF THE SOFTWARE DEMONSTRATED ON FULL-SCALE REALISTIC PROBLEMS. PARTIALLY INTEGRATE WITH EXISTING HARDWARE/SOFTWARE SYSTEMS. LIMITED DOCUMENTATION AVAILABLE. ENGINEERING FEASIBILITY FULLY DEMONSTRATED.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS.			D-TECHNOLOGY DEVELOPMENT AND PROTOTYPES TRLs 5, 6

		FOR SUBSEQUENT DEVELOPMENT PHASES.	IMPLEMENTATION DEVELOPED.		D-TECHNOLOGY DEVELOPM TRLs 5,
6	SYSTEM/SUB-SYSTEM MODEL OR PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT	A HIGH FIDELITY SYSTEM/COMPONENT PROTOTYPE THAT ADEQUATELY ADDRESSES ALL CRITICAL SCALING ISSUES IS BUILT AND OPERATED IN A RELEVANT ENVIRONMENT TO DEMONSTRATE OPERATIONS UNDER CRITICAL ENVIRONMENTAL CONDITIONS.	PROTOTYPE IMPLEMENTATION OF THE SOFTWARE DEMONSTRATED ON FULL-SCALE REALISTIC PROBLEMS. PARTIALLY INTEGRATED WITH EXISTING HARDWARE/SOFTWARE SYSTEMS. LIMITED DOCUMENTATION AVAILABLE. ENGINEERING FEASIBILITY FULLY DEMONSTRATED.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS.	E-PILOT PLAN AND SCALE UP TRLs 6, 7, 8
7	SYSTEM PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT	A HIGH FIDELITY ENGINEERING UNIT THAT ADEQUATELY ADDRESSES ALL CRITICAL SCALING ISSUES IS BUILT AND OPERATED IN A RELEVANT ENVIRONMENT TO DEMONSTRATE PERFORMANCE IN THE ACTUAL OPERATIONAL ENVIRONMENT AND PLATFORM (GROUND, AIRBORNE, OR SPACE).	PROTOTYPE SOFTWARE EXISTS HAVING ALL KEY FUNCTIONALITY AVAILABLE FOR DEMONSTRATION AND TEST. WELL INTEGRATED WITH OPERATIONAL HARDWARE/SOFTWARE SYSTEMS DEMONSTRATING OPERATIONAL FEASIBILITY. MOST SOFTWARE BUGS REMOVED. LIMITED DOCUMENTATION AVAILABLE.	DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS.	
8	ACTUAL SYSTEM COMPLETED AND "FLIGHT QUALIFIED" THROUGH TEST AND DEMONSTRATION	THE FINAL PRODUCT IN ITS FINAL CONFIGURATION IS SUCCESSFULLY DEMONSTRATED THROUGH TEST AND ANALYSIS FOR ITS INTENDED OPERATIONAL ENVIRONMENT AND PLATAFORM (GROUND, AIRBORNE, OR SPACE).	ALL SOFTWARE HAS BEEN THOROUGHLY DEBUGGED AND FULLY INTEGRATED WITH ALL OPERATIONAL HARDWARE AND SOFTWARE SYSTEMS. ALL USER DOCUMENTATION, TRAINING DOCUMENTATION, AND MAINTENANCE DOCUMENTATION COMPLETED. ALL FUNCTIONALITY SUCCESSFULLY DEMONSTRATED IN SIMULATED OPERATIONAL SCENARIOS. VERIFICATION AND VALIDATION (V&V) COMPLETED.	DOCUMENTED TEST PERFORMANCE VERIFYING AGREEMENT WITH ANALYTICAL PREDICTIONS.	
9	ACTUAL SYSTEM FLIGHT PROVEN THROUGH SUCCESSFUL MISSION OPERATIONS	THE FINAL PRODUCT IS SUCCESSFULLY OPERATED IN AN ACTUAL MISSION.	ALL SOFTWARE HAS BEEN THOROUGHLY DEBUGGED AND FULLY INTEGRATED WITH ALL OPERATIONAL HARDWARE AND SOFTWARE SYSTEMS. ALL DOCUMENTATION HAS BEEN COMPLETED. SUSTAINING SOFTWARE ENGINEERING SUPPORT IS IN PLACE. SYSTEM HAS BEEN SUCCESSFULLY OPERATED IN THE OPERATIONAL ENVIRONMENT.	DOCUMENTED MISSION OPERATIONAL RESULTS.	F-MARKET LAUNCH AND COMMERCIALIZATION TRLs 7, 8, 9

THE ACADEMIA ROLE OF KNOWLEDGE CREATION ENCOMPASSES TRL CONCENTRATIONS A AND B

A-BASIC TECHNOLOGY RESEARCH: TRL1-BASIC PRINCIPLES OBSERVED AND REPORTED; TRL2-TECHNOLOGY CONCEPT AND/OR APPLICATION; TRL3-ANALYTICAL AND EXPERIMENTAL CRITICAL FUNCTION AND/OR CHARACTERISTIC PROOF OF CONCEPT.

B-RESEARCH TO PROVE FEASIBILITY: TRL2-; TRL3-; TRL4-COMPONENT AND/OR BREADBOARD VALIDATION IN LABORATORY ENVIRONMENT).

TRL OVERLAYS: CONCENTRATIONS A AND B: TRL2 AND TRL3.

THE GOVERNMENT AND CIVIL SOCIETY REPRESENTATIVE ENTITIES ROLE OF INCENTIVE AND SUPPORT TO COLLABORATION ENCOMPASSES TRL CONCENTRATIONS C AND D

C-TECHNOLOGY DEMONSTRATION: TRL3-; TRL4-; TRL5-COMPONENT AND/OR BREADBOARD VALIDATION IN RELEVANT ENVIRONMENT.

D-TECHNOLOGY DEVELOPMENT AND PROTOTYPE: TRL5-; TRL6- SYSTEM/SUB-SYSTEM MODEL OR PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT.

TRL OVERLAYS: (I) CONCENTRATIONS C AND D: TRL5; (II) CONCENTRATIONS C AND B: TRL3 AND TRL4; (III) CONCENTRATIONS C AND A: TRL3.

THE PRIVATE SECTOR, STATE ENTITIES AND NGOs ROLE OF COMMERCIAL PRODUCTS AND PUBLIC SERVICES IMPLEMENTATION SUCCESS ENCOMPASSES TRL CONCENTRATIONS E AND F

E-PILOT PLAN AND SCALE UP: TRL6-; TRL7-SYSTEM PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT; TRL8-ACTUAL SYSTEM COMPLETED AND "FLIGHT QUALIFIED" THROUGH TEST AND DEMONSTRATION.

F-MARKET LAUNCH AND COMMERCIALIZATION: TRL7-; TRL8-; TRL9-ACTUAL SYSTEM FLIGHT PROVEN THROUGH SUCCESSFUL MISSION OPERATIONS).

TRL OVERLAYS: (I) CONCENTRATIONS E AND F: TRL7 AND TRL8; (II) CONCENTRATIONS E AND D: TRL6.

NASA TECHNOLOGY READINESS LEVELS-TRLS

TOMANDO OS CRITÉRIOS DE SAÍDA DE TRL/TRL EXIT CRITERIA COMO BASE, A FIGURA MOSTRA AS POSSÍVEIS ENTREGAS DOS PRINCIPAIS PARTICIPANTES VINCULADAS ÀS CONCENTRAÇÕES DE TRLS:

ACADEMIA

A. TRL1-PEER REVIEWED PUBLICATION OF RESEARCH UNDERLYING THE PROPOSED CONCEPT/APPLICATION;

TRL2-DOCUMENTED DESCRIPTION OF THE APPLICATION/CONCEPT THAT ADDESSES FEASIBILITY AND BENEFIT;

TRL3-DOCUMENTED ANALYTICAL/EXPERIMENTAL RESULTS VALIDATING PREDICTIONS OF KEY PARAMETERS.

B. TRL2-; TRL3-; TRL4-DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS. DOCUMENTED DEFINITION OF RELEVANT ENVIRONMENT.

GOVERNMENT AND CIVIL SOCIETY REPRESENTATIVE ENTITIES

C. TRL3-; TRL4-;

D. TRL5-DOCUMENTED TEST PERFORMANCE DEMONSTRATING AGREEMENT WITH ANALYTICAL PREDICTIONS. DOCUMENTED DEFINITION OF SCALING REQUIREMENTS.

TRL5-; TRL6-SYSTEM/SUB-SYSTEM MODEL OR PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT.

PRIVATE SECTOR, STATE ENTITIES AND NGOs

E. TRL6-;

TRL7-SYSTEM PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT;

TRL8-DOCUMENTED TEST PERFORMANCE VERIFYING AGREEMENT WITH ANALYTICAL PREDICTIONS.

F. TRL7-; TRL8-;

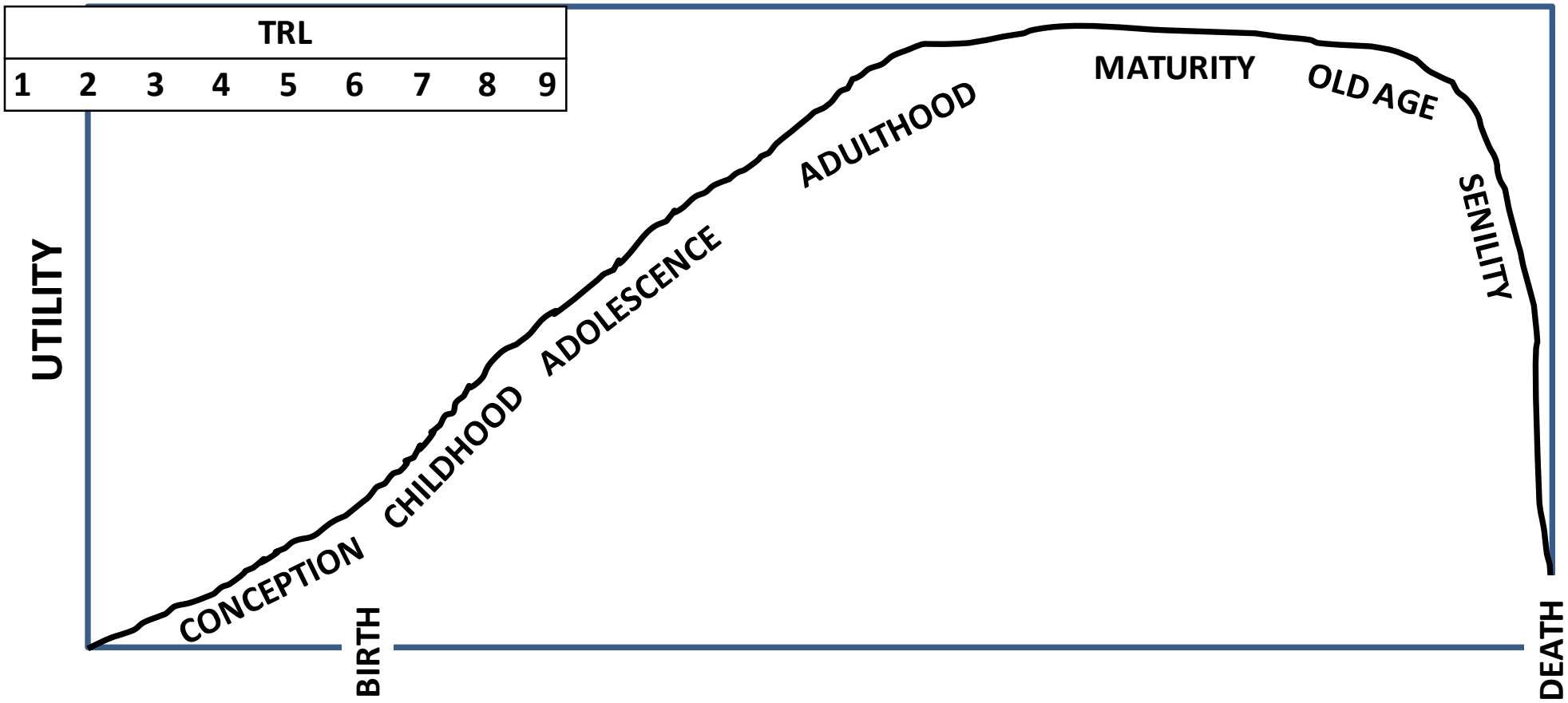
TRL9-DOCUMENTED MISSION OPERATIONAL RESULTS.

DIAGRAMA BALEIA

NO INÍCIO DE PALESTRA SOBRE CICLO DE VIDA DE TECNOLOGIA/TECHNOLOGY LIFE CYCLE (https://apps.dtic.mil/dtic/tr/fulltext/u2/a509619.pdf), BILL NOLTE (NOLTE, 2006) MOSTRA QUE EXISTEM DIFERENÇAS E SIMILARIDADES ENTRE OS CICLOS DE VIDA DE PROJETOS DOD E NASA, APESAR DE AMBOS INCLUIREM OS 9 TRLS. NO QUE SE REFERE A DIFERENÇAS, O AUTOR DIZ QUE "MOST NASA PROJECTS DO NOT ENTER INTO MULTIPLE ITEM PRODUCTION CYCLES" E "MOST NASA PROJECTS ARE NOT AVAILABLE FOR RETROFITTING/REPAIR". QUANTO A SIMILARIDADES, DOD E NASA AFIRMA QUE "PROJECT COST, SCHEDULE, AND PERFORMANCE RISK ARE MOST LIKELY SIGNIFICANTLY REDUCED IF TECHNOLOGIES ARE WELL DEVELOPED PRIOR TO LEVEL/MILESTONE C".

EM SEGUIDA É APRESENTADO UM DIAGRAMA EM FORMATO DE BALEIA, DESTACANDO 9 FASES DO TEMPO DE VIDA (CONCEPÇÃO/CONCEPTION, NASCIMENTO/BIRTH, INFÂNCIA/CHILDHOOD, ADOLESCÊNCIA/ADOLESCENCE, IDADE ADULTA/ADULTHOOD, MATURIDADE/MATURITY, VELHICE/OLD AGE, SENILIDADE/SENILITY, MORTE/DEATH) NO EIXO HORIZONTAL E UTILITY NO EIXO VERTICAL ("DIAGRAMA BALEIA/WHALE CHART") COMO "UMA METÁFORA BIOLÓGICA PARA O CICLO DE VIDA DE TECNOLOGIA". NOLTE ENTÃO PONDERA QUE "A TECHNOLOGY'S USEFULNESS CHANGES OVER TIME: UTILITY INCREASES AS A TECHNOLOGY MATURES. UTILITY DECREASES AS A TECHNOLOGY BECOMES OBSOLETE."

COM BASE NO DIAGRAMA BALEIA DA FIGURA S1.4.15, NOLTE CONCLUI QUE "TECHNOLOGY READINESS LEVELS OCCUR EARLY IN THE TECHNOLOGY LIFE CYCLE", COM TRL9 PRÓXIMO DO FIM DA ADOLESCÊNCIA E INÍCIO DA IDADE ADULTA, QUANDO/ONDE SE INICIA A PRODUÇÃO, DESDOBRAMENTO, OPERAÇÃO E SUPORTE, DURANTE A IDADE ADULTA/ADULTHOOD, MATURIDADE/MATURITY, VELHICE/OLD AGE, SENILIDADE/SENILITY, MORTE/DEATH. COM BASE NO DIAGRAMA BALEIA É POSSÍVEL IR ALÉM DOS NÍVEIS TRL, LEVANDO EM CONSIDERAÇÃO NÍVEIS DE MANUFATURA, INTEGRAÇÃO, SISTEMAS E COMERCIAL, EM TERMOS DE CICLO DE VIDA DE TECNOLOGIA MAIS LONGO.


TRL
1 2 3 4 5 6 7 8 9
UTILITY
CONCEPTION
CHILDHOOD
ADOLESCENCE
ADULTHOOD
MATURITY
OLD AGE
SENILITY
BIRTH
DEATH

DETALHAMENTO DE PRONTIDÃO EM INOVAÇÃO

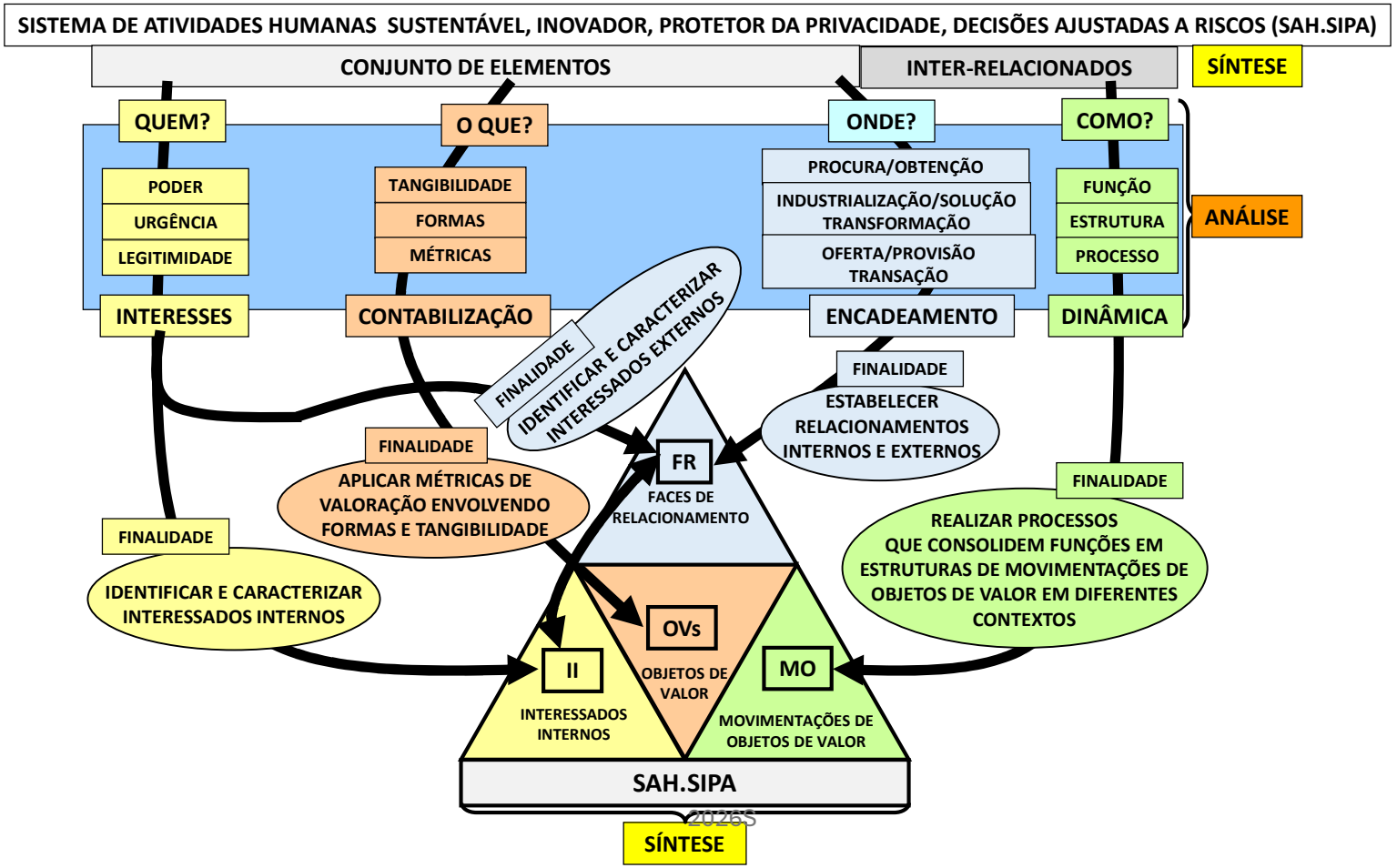
AS PRINCIPAIS REFERÊNCIAS PARA PRONTIDÃO EM INOVAÇÃO SÃO: SYSTEMS ENGINEERING GUIDEBOOK, DEPARTMENT OF DEFENSE, UNITED STATES OF AMERICA, FEBRUARY, 2022, (https://ac.cto.mil/wp-content/uploads/2022/02/systems-eng-guidebook_feb2022-cleared-slp.pdf), TECHNOLOGY READINESS ASSESSMENT GUIDE, U. S. GOVERNMENT ACCOUNTABILITY OFFICE-GAO, 2020 (https://www.gao.gov/assets/gao-20-48g.pdf) E O SYSTEM READINESS ASSESSMENT (SRA) ENGINEERING HANDBOOK, VERSION 2.1, 2019.

A PRONTIDÃO EM INOVAÇÃO É A SÍNTESE DE COMPONENTES DE PRODUTO, PROCESSO E MERCADO.

A DIMENSÃO PRONTIDÃO (READINESS) EM INOVAÇÃO ESTÁ CONCEITUALMENTE ASSOCIADA AOS 5 PRIMEIROS TRLS DA NASA: TRL1-BASIC PRINCIPLES OBSERVED AND REPORTED; TRL2-TECHNOLOGY CONCEPT AND/OR APPLICATION; TRL3-ANALYTICAL AND EXPERIMENTAL CRITICAL FUNCTION AND/OR PROOF OF CONCEPT; TRL4-COMPONENT AND/OR BREADBOARD VALIDATION IN LABORATORY ENVIRONMENT; AND TRL5-COMPONENT AND/OR BREADBOARD VALIDATION IN RELEVANT ENVIRONMENT.

VALE RESSALTAR DUAS CONCENTRAÇÕES DE 3 TRLS EM ACADEMIA: KNOWLEDGE CREATION (A-BASIC TECHNOLOGY RESEARCH: TRLS 1,2,3; E B-RESEARCH TO PROVE FEASIBILITY: TRLS 2,3,4), COMO TAMBÉM UMA CONCENTRAÇÃO EM GOVERNMENT AND CIVIL SOCIETY REPRESENTATIVE ENTITIES: INCENTIVE AND SUPPORT TO COLLABORATION (C-TECHNOLOGY DEMONSTRATION: TRLS 3,4,5).

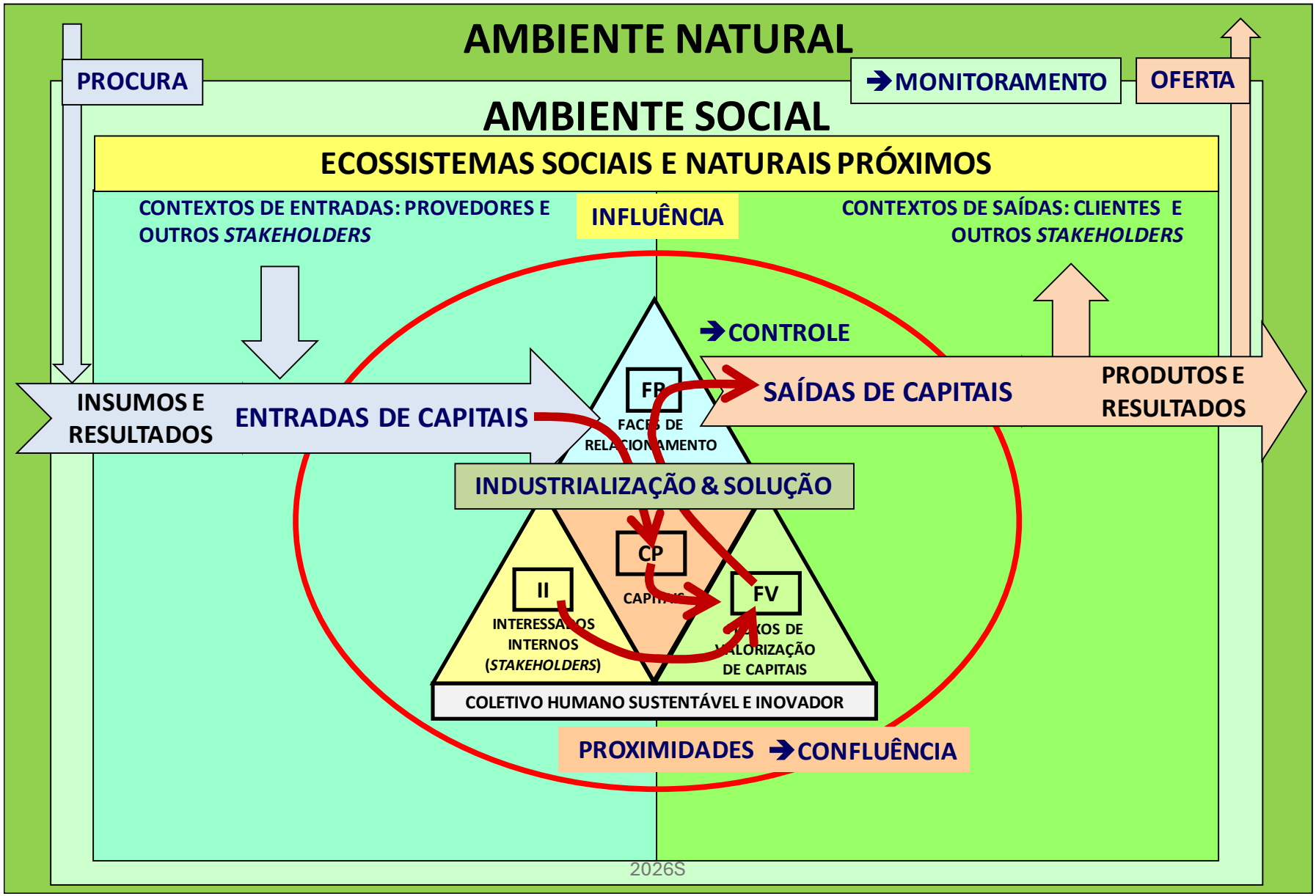
TOMANDO COMO PREMISSA QUE PRONTIDÃO EM INOVAÇÃO ESTÁ DIRETAMENTE ASSOCIADA A SISTEMAS DE ATIVIDADES HUMANAS (SAH) QUE SEJAM SUSTENTÁVEIS, INOVADORES, PROTETORES DA PRIVACIDADE, COM DECISÕES AJUSTADAS A RISCOS (SAH.SIPA), CABE DESCREVER A ARQUITETURA DE UM SAH.SIPA EM TERMOS DE (i) STAKEHOLDERS INTERNOS E EXTERNOS (QUEM); (ii) SEGMENTAÇÃO DE OBJETOS QUE CRIAM VALOR (CAPITAIS/OBJETOS DE VALOR – OVs: O QUE); (iii) ESPAÇOS DE RELACIONAMENTOS INTERNOS E EXTERNOS (ONDE); E (iv) FLUXOS DE CAPITAIS/OVs (COMO), CONFORME A FIGURA.



A FIGURA MOSTRA OS AMBIENTES NATURAL E SOCIAL, BEM COMO OS MACROSSISTEMAS SOCIAIS COM INTERESSES DELIMITANDO SUAS MACROFRONTEIRAS, ALÉM DOS CONTEXTOS ESPECÍFICOS QUE, JUNTAMENTE COM AS BORDAS E O ESCOPO CONFIGURAM A COESÃO DO SSIPA.

AS SETAS DA PARTE CENTRAL DA FIGURA EXPLICITAM OS RELACIONAMENTOS ASSOCIADOS A SEUS QUATRO CONSTITUINTES:

- **RELACIONAMENTOS EXTERNOS**
  - **ENTRADAS DE CAPITAIS DE CONTEXTOS DE ENTRADAS, COMPOSTOS POR PROVEDORES E OUTROS INTERESSADOS EXTERNOS;**
  - **SAÍDAS DE CAPITAIS PARA CONTEXTOS DE SAÍDAS, COMPOSTOS POR CLIENTES E OUTROS INTERESSADOS EXTERNOS.**
- **RELACIONAMENTOS INTERNOS**
- **OS CAPITAIS QUE ENTRAM VIA ENTRADA DE CAPITAIS SÃO COMBINADOS COM AS DISPONIBILIDADES INTERNAS, VISANDO À ADIÇÃO INTERNA DE VALOR E À ALOCAÇÃO DO VALOR TOTAL ADICIONADO;**
- **OS *STAKEHOLDERS* INTERNOS INTERAGEM DE ACORDO COM OS SEUS INTERESSES EM FLUXOS DE CAPITAIS QUE RESULTAM EM ADIÇÃO INTERNA DE VALOR E À ALOCAÇÃO DO VALOR TOTAL ADICIONADO. VALE RESSALTAR QUE ENTRE OS *STAKEHOLDERS* INTERNOS OCORREM INTERAÇÕES EM TRÊS NÍVEIS DE CONVIVÊNCIA: PESSOAL, INSTITUCIONAL E EM REDE.**
- **OS CAPITAIS RESULTANTES DA VALORIZAÇÃO INTERNA SÃO ENCAMINHADOS PARA AS SAÍDAS DE CAPITAIS.**


AMBIENTE NATURAL
PROCURA
➔MONITORAMENTO
OFERTA
AMBIENTE SOCIAL
ECOSSISTEMAS SOCIAIS E NATURAIS PRÓXIMOS
CONTEXTOS DE ENTRADAS: PROVEDORES E OUTROS *STAKEHOLDERS*
INFLUÊNCIA
CONTEXTOS DE SAÍDAS: CLIENTES E OUTROS *STAKEHOLDERS*
➔CONTROLE
INSUMOS E RESULTADOS
ENTRADAS DE CAPITAIS
SAÍDAS DE CAPITAIS
PRODUTOS E RESULTADOS
FR
FACES DE RELACIONAMENTO
INDUSTRIALIZAÇÃO & SOLUÇÃO
CP
II
INTERESSADOS INTERNOS (*STAKEHOLDERS*)
FV
VALORIZAÇÃO DE CAPITAIS
COLETIVO HUMANO SUSTENTÁVEL E INOVADOR
PROXIMIDADES ➔CONFLUÊNCIA

COMO 4 TÍTULOS GERAM MAIORES POSSIBILIDADES DE INTERSEÇÕES PARA AS BUSCAS BIBLIOMÉTRICAS, AO INVÉS DE CONSOLIDAR PRONTIDÃO EM INOVAÇÃO EM 3 TÍTULOS (PRODUTO, PROCESSO, MERCADO), UMA POSSIBILIDADE É LEVAR EM CONTA OS 2 PRINCIPAIS PROCESSOS (CUSTOMER RELATIONSHIP MANAGEMENT – CRM E SUPPLIER RELATIONSHIP MANAGEMENT – SRM) VINCULADOS A MERCADOS DE OBTENÇÃO/PROCURA E PROVISÃO/OFERTA DA GESTÃO DA CADEIA DE SUPRIMENTO (SUPPLY CHAIN MANAGEMENT – SCM) E DIVIDIR MERCADO EM 2 COMPONENTES (OBTENÇÃO E PROVISÃO), CRIANDO O MODELO PISO, EM TERMOS DE 4 GRANDES TÍTULOS: (i) P-PROCURA/OBTENÇÃO; (ii) I-INDUSTRIALIZAÇÕ/PROCESSO; (iii) S-SOLUÇÃO/PRODUTO; E (iv) O-OFERTA/PROVISÃO.

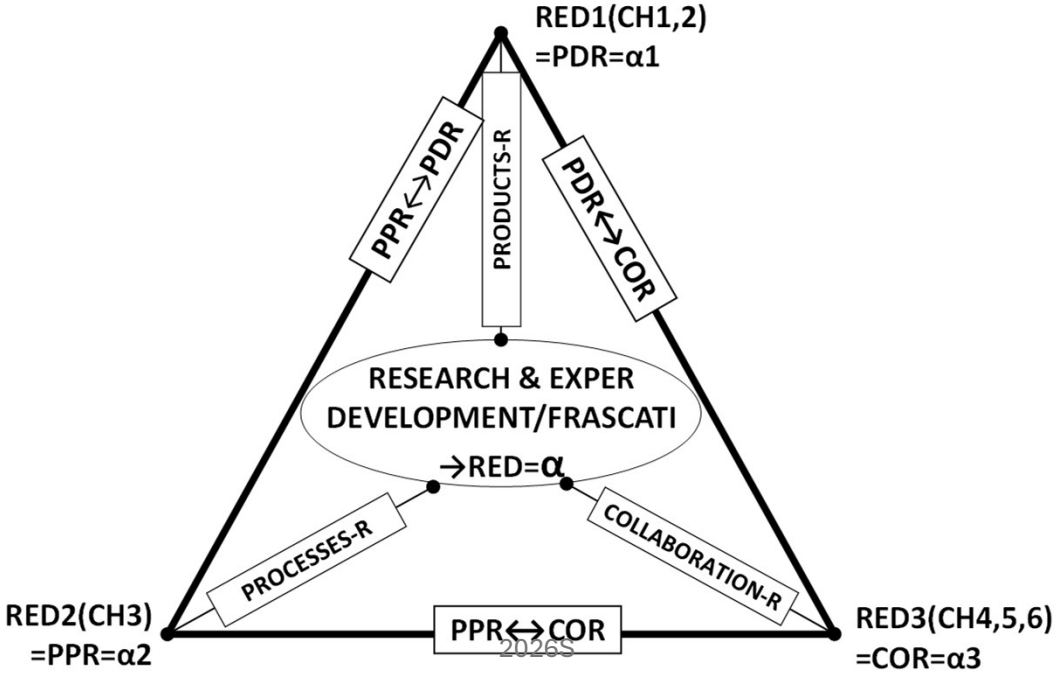
MODELO BUSCAS BIBLIOMÉTRICAS: PRONTIDÃO EM INOVAÇÃO

A ABORDAGEM DE TRIÂNGULOS E TETRAEDROS UTILIZADA NO TÓPICO DESENVOLVIMENTO SUSTENTÁVEL: MODELO DE TRIÂNGULOS E TETRAEDROS É AQUI REPETIDA PARA MODELAR AS BUSCAS BIBLIOMÉTRICAS DE PRONTIDÃO EM INOVAÇÃO.

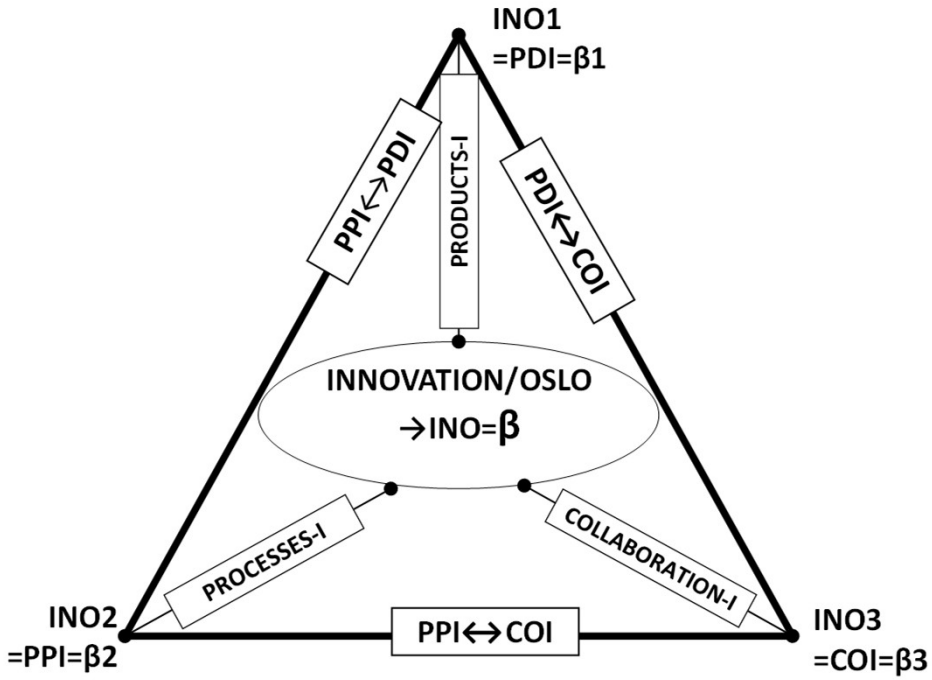
O PRIMEIRO PASSO É CONSOLIDAR AS DIMENSÕES DE PRONTIDÃO EM INOVAÇÃO EM 4 CONCENTRAÇÕES, DENOTADAS PELAS LETRAS GREGAS α, β, γ, δ: (i) PESQUISA E DESENVOLVIMENTO EXPERIMENTAL=RESEARCH AND EXPERIMENTAL DEVELOPMENT=RED=α; (ii) INOVAÇÃO=INNOVATION=INV=β; (iii) NÍVEIS DE PRONTIDÃO=READINESS LEVELS=RDL=γ; e (iv) AVALIAÇÃO DE PRONTIDÃO=READINESS ASSESSMENT=RDA=δ. O QUADRO DISCRIMINA OS COMPONENTES DA PRONTIDÃO EM INOVAÇÃO EM TERMOS DESSAS 4 CONCENTRAÇÕES.

INNOVATION READINESS→FRASCATI/OSLO/READINESS LEVELS/READINESS LEVELS ASSESSMENT CONCENTRATIONS		
RESEARCH AND EXPERIMENTAL DEVELOPMENT/FRASCATI MANUAL: RED=α		
PRODUCTS-R: PDR=RED1=α1		
PROCESSES-R: PPR=RED2=α2		
COLLABORATION-R: COR=RED3=α3		
INNOVATION/OSLO MANUAL: INO=β	READINESS LEVELS: RLV=γ	READINESS LEVELS ASSESSMENT: RLA=δ
PRODUCTS-I: PDI=INO1=β1	TECHNOLOGY&MANUFACTURING: TML=RLV1=γ1	TECHNOLOGY&MANUFACTURING: ATM=RLA1=δ1
PROCESSES-I: PPI=INO2=β2	SYSTEM&INTEGRATION: SIL=RLV2=γ2	SYSTEM&INTEGRATION: ASI=RLA2=δ2
COLLABORATION-I: COI=INO3=β3	COMPONENT: COL=RLV3=γ3	COMPONENT: ACO=RLA3=δ3

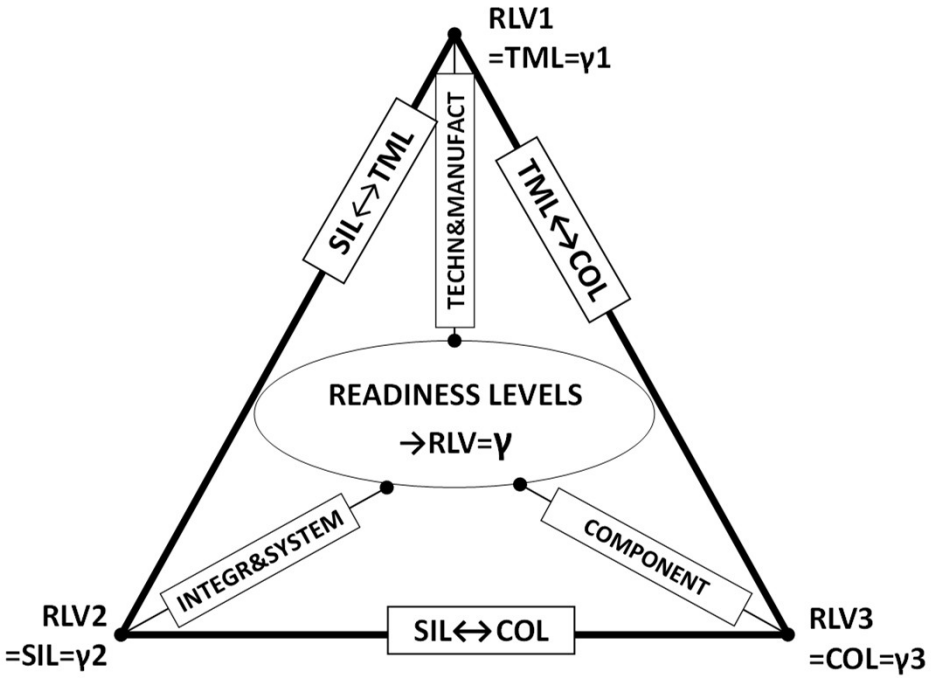
COM FOCO NO FRASCATI MANUAL 2015, DA OECD (https://www.oecd.org/en/publications/2015/10/frascati-manual-2015_g1g57dcb.html), O TRIÂNGULO DA FIGURA MOSTRA A CONSOLIDAÇÃO DE PESQUISA E DESENVOLVIMENTO EXPERIMENTAL=RESEARCH AND EXPERIMENTAL DEVELOPMENT=RED=A EM TERMOS DE PESQUISA E DESENVOLVIMENTO EXPERIMENTAL EM PRODUTOS=PRODUCTS/R=PDR=RED1=α1; PESQUISA E DESENVOLVIMENTO EXPERIMENTAL EM PROCESSOS=PROCESSES/R=PPR=RED2=α2; COLABORAÇÃO EM PESQUISA E DESENVOLVIMENTO=COLLABORATION/R=COR=RED3=α3. ALÉM DE CONFIGURAR RESEARCH AND EXPERIMENTAL DEVELOPMENT=RED=α COMO SÍNTESE DOS 3 COMPONENTES EM SUA FACE, AS ARESTAS DO TRIÂNGULO DA FIGURA S1.4.62 MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE SEUS COMPONENTES. VALE RESSALTAR QUE NO TRIÂNGULO DA FIGURA, A DIMENSÃO PRODUCTS/R=RED=α APARECE NA FACE DO TRIÂNGULO E AS ARESTAS DO TRIÂNGULO MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE ESSES COMPONENTES.



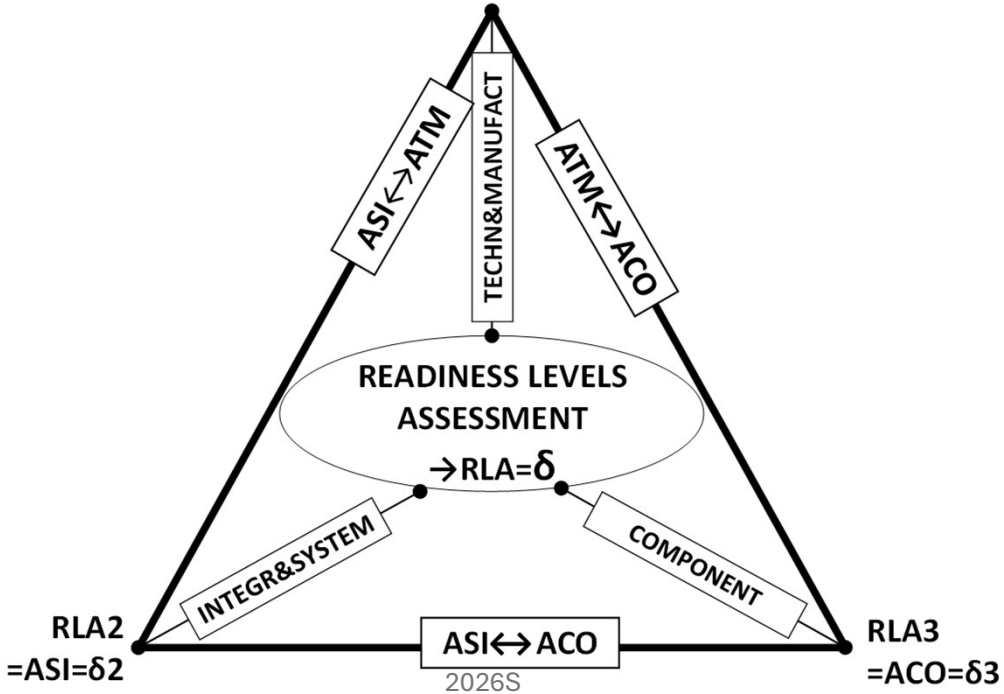
O TRIÂNGULO DA **FIGURA** MOSTRA A CONSOLIDAÇÃO DE **INOVAÇÃO=INNOVATION=INO=β** EM TERMOS DE **INOVAÇÃO EM PRODUTOS=PRODUCTS/I=PDI=INO1=β1**; **INOVAÇÃO EM PROCESSOS=PROCESSES/IN=PPI=INO2=β2**; **COLABORAÇÃO=COLLABORATION/I=COI=INO3=β3**. Assim como no triângulo da **Figura "RESEARCH&EXPERIMENTAL DEV/FRASCATI"**, A DIMENSÃO **INNOVATION=INO=β** É A FACE DO TRIÂNGULO E AS ARESTAS DO TRIÂNGULO MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE ESSES COMPONENTES.



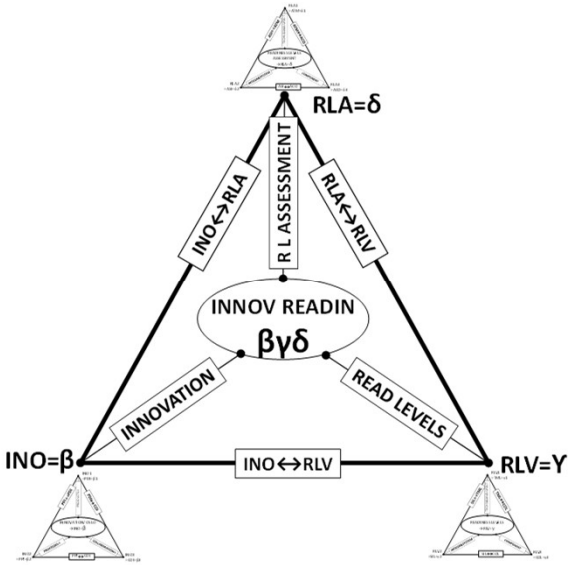
O TRIÂNGULO DA **FIGURA** MOSTRA A CONSOLIDAÇÃO DE **NÍVEIS DE PRONTIDÃO=READINESS LEVELS=RLV** EM TERMOS DE **NÍVEIS DE PRONTIDÃO TECNOLÓGICA** E **NÍVEIS DE PRONTIDÃO DE FABRICAÇÃO=TECHNOLOGY READINESS LEVELS** AND **MANUFACTURING READINESS LEVELS=TML=RLV1=γ1**; **NÍVEIS DE PRONTIDÃO DE INTEGRAÇÃO** E **NÍVEIS DE PRONTIDÃO DE SISTEMAS=INTEGRATION READINESS LEVELS** AND **SYSTEMS READINESS LEVELS=SIL=RLV2=γ2**; **NÍVEIS DE PRONTIDÃO COMERCIAL/COMPONENTE=COMMERCIAL/COMPONENT READINESS LEVELS=COL=RLV3=γ3**. ASSIM COMO NOS TRIÂNGULOS DAS **FIGURAS "RESEARCH&EXPERIMENTAL DEV/FRASCATI"** E **"INNOVATION/OSLO"**, A DIMENSÃO **READINESS LEVELS=RDL=γ** APARECE NA FACE DO TRIÂNGULO E AS ARESTAS DO TRIÂNGULO MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE ESSES COMPONENTES.



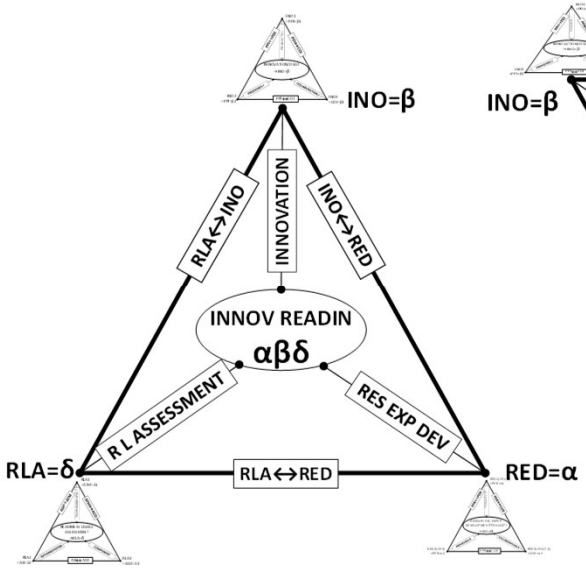
O TRIÂNGULO DA **FIGURA** MOSTRA A CONSOLIDAÇÃO DE **AVALIAÇÃO DE NÍVEIS DE PRONTIDÃO=READINESS LEVEL ASSESSMENT=RLA** EM TERMOS DE **AVALIAÇÃO DE NÍVEIS DE PRONTIDÃO TECNOLÓGICA** E **AVALIAÇÃO DE NÍVEIS DE PRONTIDÃO DE FABRICAÇÃO=TECHNOLOGY READINESS LEVEL ASSESSMENT** AND **MANUFACTURING READINESS LEVEL ASSESSMENT=ATM=RLA1=δ1**; **AVALIAÇÃO DE NÍVEIS DE PRONTIDÃO DE INTEGRAÇÃO** E **AVALIAÇÃO DE NÍVEIS DE PRONTIDÃO DE SISTEMAS=INTEGRATION READINESS LEVEL ASSESSMENT** AND **SYSTEMS READINESS LEVEL ASSESSMENT=ASI=RLA2=δ2**; **AVALIAÇÃO DE NÍVEIS DE PRONTIDÃO COMERCIAL/COMPONENTE=COMMERCIAL/COMPONENT READINESS LEVEL ASSESSMENT=ACO=RLA3=δ3**. ASSIM COMO NOS TRIÂNGULOS DAS **FIGURAS "RESEARCH&EXPERIMENTAL DEV/FRASCATI", "INNOVATION/OSLO" E "READINESS LEVELS"**, A DIMENSÃO **READINESS LEVEL ASSESSMENT=RLA=δ** APARECE NA FACE DO TRIÂNGULO E AS ARESTAS DO TRIÂNGULO MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE ESSES COMPONENTES.



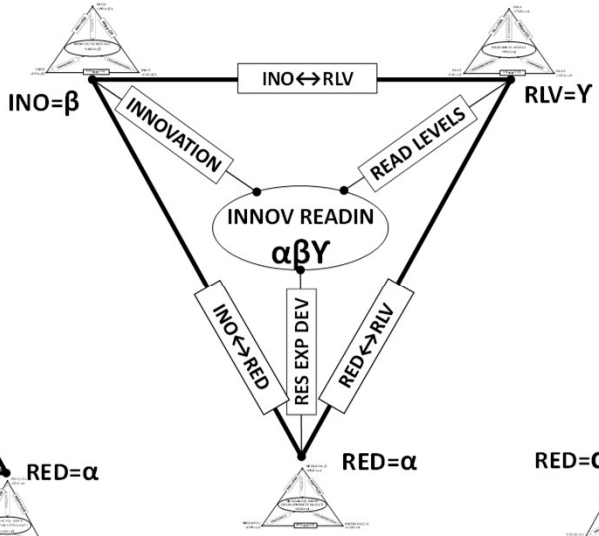
CADA UM DOS TRIÂNGULOS DAS **FIGURAS** EXPLICITAM A SÍNTESE DE 3 DAS 4 **CONCENTRAÇÕES DE PRONTIDÃO EM INOVAÇÃO** (**PESQUISA E DESENVOLVIMENTO EXPERIMENTAL=RED=α**; **INOVAÇÃO=INNOVATION=INO=β**; **NÍVEIS DE PRONTIDÃO=READINESS LEVEL=RLV=γ**; E **AVALIAÇÃO DE NÍVEIS DE PRONTIDÃO=READINESS LEVEL ASSESSMENT=RLA=δ**, EM QUE OS TRIÂNGULOS EM SEUS VÉRTICES REFEREM-SE AOS RESPECTIVOS COMPONENTES DAS FIGURAS **"RESEARCH&EXPERIMENTAL DEV/FRASCATI", "INNOVATION/OSLO", "READINESS LEVELS" E "READINESS LEVELS ASSESSMENT".**



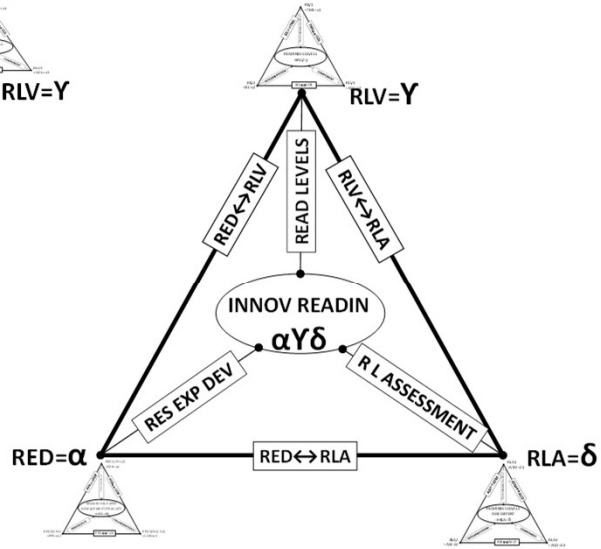
TRIÂNGULO SÍNTESE βγδ PRONTIDÃO EM INOVAÇÃO



TRIÂNGULO SÍNTESE αβδ PRONTIDÃO EM INOVAÇÃO



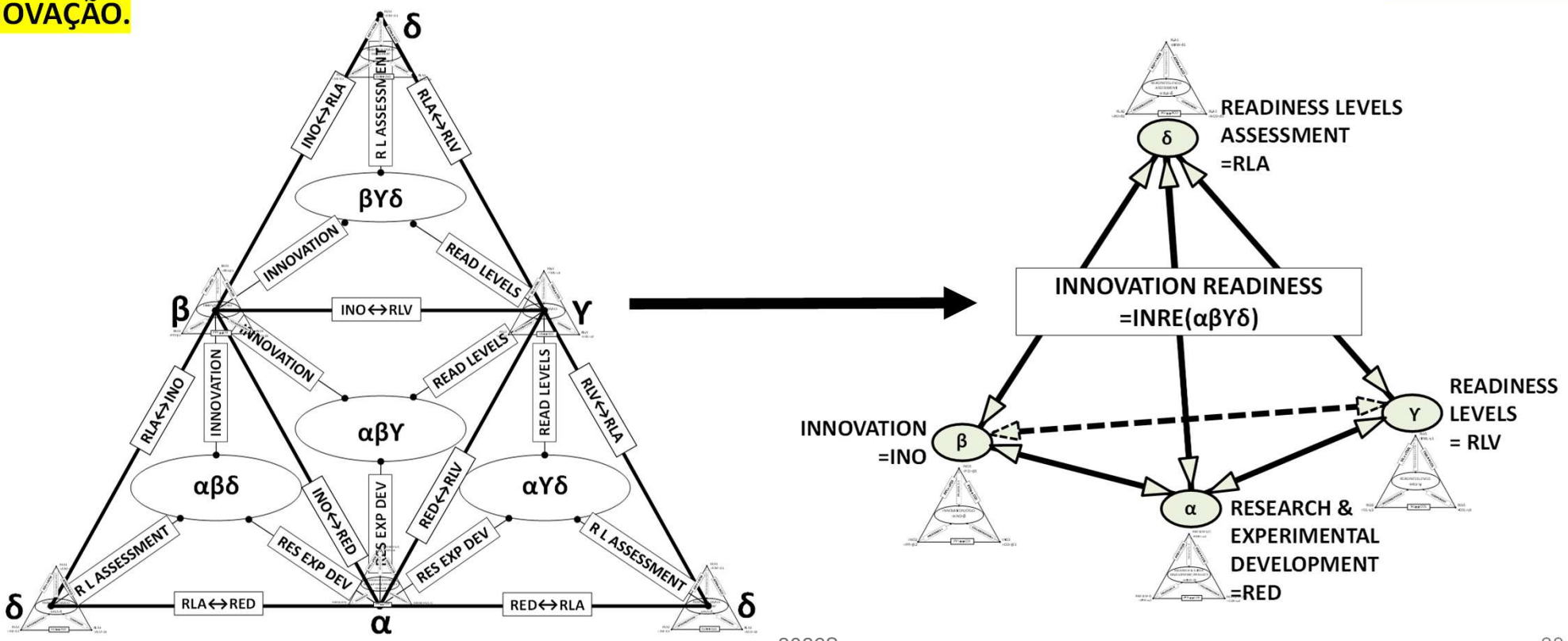
TRIÂNGULO SÍNTESE αβγ PRONTIDÃO EM INOVAÇÃO



TRIÂNGULO SÍNTESE αγδ PRONTIDÃO EM INOVAÇÃO

OBTÉM-SE O TRIÂNGULO DA FIGURA À ESQUERDA, QUE CONSTITUI UM TETRAEDRO PLANIFICADO, CONSTITUÍDO PELOS TRIÂNGULOS SÍNTESE (α,β,δ), (α,β,γ), (α,γ,δ) e (β,γ,δ).

O RESULTADO ESPERADO DO PROCESSO DE PREENCHIMENTO, VIA BIBLIOMETRIA, DO TETRAEDRO PLANIFICADO DA FIGURA À ESQUERDA É A SUA EFETIVA EVOLUÇÃO PARA UM TETRAEDRO ESPACIAL, COMO MOSTRA A FIGURA À DIREITA, QUE SINTETIZA AS BUSCAS BIBLIOMÉTRICAS ENVOLVENDO "RESEARCH&EXPERIMENTAL DEV/FRASCATI", "INNOVATION/OSLO", "READINESS LEVELS" E "READINESS LEVEL ASSESSMENT" RELATIVAS À PRONTIDÃO EM INOVAÇÃO.



MODELO BUSCAS BIBLIOMÉTRICAS: CONSOLIDA SUSTENTABILIDADE/INOVAÇÃO/PRIVACIDADE

COM BASE NOS TÓPICOS DESENVOLVIMENTO SUSTENTÁVEL (ARQUIVO 03.01IpgSipaSUSTENTABILIDADE...), PRONTIDÃO EM INOVAÇÃO (ARQUIVO 03.02IpgSipaINOVAÇÃO...) E PROTEÇÃO À PRIVACIDADE (ARQUIVO 03.03IpgSipaPRIVACIDADE...), PODE-SE CONSTRUIR O TRIÂNGULO DA FIGURA, QUE CONSOLIDA OS RELACIONAMENTOS SUSTENTABILIDADE, INOVAÇÃO E PRIVACIDADE EM UM TRIÂNGULO.

NOTAR AS CÓPIAS REDUZIDAS NOS VÉRTICES DO TRIÂNGULO DA FIGURA ASSOCIADAS À RESPECTIVA DIMENSÃO (SUSTENTABILIDADE, INOVAÇÃO E PRIVACIDADE).

